$ToTpower := 5000$

$Voltage := 4160$

$arccos := \cos^{-1}$

$pf := .9$

$$I := \frac{ToTpower}{Voltage} \cdot arccos(pf)$$

$$Z := \frac{Voltage}{I}$$

$$Vdrop := \frac{1}{2} \cdot Z \cdot \frac{5000}{5280} \cdot I \cdot \left(1 + \frac{1}{10}\right)$$

$$Vdrop = 2.167 \cdot 10^3$$

$$Vdroppercent := \frac{Vdrop}{\frac{Voltage}{\sqrt{3}}} \cdot 100$$

$$Vdroppercent = 90.211$$

$$Ploss := 3 \left(\frac{1}{3} \cdot Z \cdot \left(I^2\right) \cdot \frac{1}{3} + \frac{1}{2 \cdot 10} + \frac{1}{6 \cdot 100}\right)$$

$$Ploss = 751.866$$

$$Vdrop2 := \frac{1}{2} \cdot \frac{Z}{2} \cdot \frac{5000}{5280} \cdot I \cdot \left(1 + \frac{1}{\frac{10}{2}}\right)$$

$$Vdrop2 = 1.182 \cdot 10^3$$

$$Vdroppercent2 := \frac{Vdrop2}{\frac{Voltage}{\sqrt{3}}} \cdot 100$$

$$Vdroppercent2 = 49.206$$

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